Lesson 2.7 — Complex Numbers

Big idea: When the discriminant goes negative, real numbers can't finish the problem. Complex numbers extend the real line by introducing $i = \sqrt{-1}$.

Quick review

Definition: $i = \sqrt{-1}$, so $i^2 = -1$.
Powers of i cycle: $i, -1, -i, 1$, then repeat.
Add/subtract: combine real parts and imaginary parts separately. $(a + bi) + (c + di) = (a + c) + (b + d)i$.
Multiply: use FOIL and replace i^2 with -1.
Complex conjugate: the conjugate of $a + bi$ is $a - bi$. $(a + bi)(a - bi) = a^2 + b^2$ (always real).

Worked example

Worked example. Simplify $(3 + 2i)(4 - i)$.

$= 12 - 3i + 8i - 2i^2 = 12 + 5i - 2(-1) = 14 + 5i$.

Warm-ups

1. Simplify $\sqrt{-16}$.
2. Simplify $(2 + 3i) + (5 - i)$.
3. Simplify i^3.

Practice

1. Simplify $(4 - i) - (2 + 3i)$.
2. Simplify $(1 + 2i)(3 - i)$.
3. Simplify $(5 + 2i)(5 - 2i)$.

Challenge

1. Solve $x^2 + 6x + 13 = 0$ over the complex numbers.
2. Simplify $(2 + i)/(1 - i)$. (Hint: multiply numerator and denominator by the conjugate of the denominator.)

Answer key — Lesson 2.7

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $4i$.
2. $7 + 2i$.
3. $i^3 = i^2 \cdot i = -i$.

Practice

1. $2 - 4i$.
2. $3 - i + 6i - 2i^2 = 3 + 5i + 2 = 5 + 5i$.
3. $25 - 4i^2 = 25 + 4 = 29$.

Challenge

1. $x = (-6 \pm \sqrt{(36 - 52)})/2 = (-6 \pm 4i)/2 = -3 \pm 2i$.
2. Multiply by $(1 + i)/(1 + i)$: $[(2 + i)(1 + i)] / [(1 - i)(1 + i)] = (2 + 2i + i + i^2)/(1 + 1) = (1 + 3i)/2 = 1/2 + (3/2)i$.